

An Exact Fifteen-Dimensional Local Koopman Jet for a Polynomial Hénon Germ

Anonymous Route-A report

Abstract

We construct the exact Koopman pullback of the polynomial germ $F(u, v) = (u^2 + 3u/2 - v/2, u)$ on the local algebra $A_4 = \mathbb{Q}[u, v]/(u, v)^5$. In its fifteen-element monomial basis, the operator has trace $129/16$, determinant 2^{-20} , and a characteristic polynomial determined by five associated-graded blocks. The nonlinear term changes eleven matrix entries, but only in strictly degree-raising positions; the correction is nilpotent of index four. The resulting finite determinant is an exact operator-first Route-A prefix. It is not a global Koopman spectrum or Fredholm determinant, and no arithmetic or Route-B claim is made.

1 A finite local operator

Let $\mathfrak{m} = (u, v)$ and freeze

$$F(u, v) = \left(u^2 + \frac{3}{2}u - \frac{1}{2}v, u \right), \quad F(0, 0) = (0, 0).$$

Because both coordinates of F lie in $\mathfrak{m}$, pullback sends $\mathfrak{m}^5$ into itself. Composition therefore defines the rational operator

$$K : A_4 \longrightarrow A_4, \quad K[p] = [p \circ F], \quad A_4 = \mathbb{Q}[u, v]/\mathfrak{m}^5.$$

We order the basis by total degree,

$$\mathcal{B} = (1; u, v; u^2, uv, v^2; u^3, u^2v, uv^2, v^3; u^4, u^3v, u^2v^2, uv^3, v^4),$$

and use the convention that column j contains $K(\mathcal{B}_j)$. Exact truncated substitution gives all 225 matrix cells in the accompanying evidence. For example,

$$K[u] = u^2 + \frac{3}{2}u - \frac{1}{2}v, \quad K[v] = u,$$

so the finite operator already retains nonlinear information.

2 Graded blocks and exact determinant

The total-degree filtration is K -invariant. On its degree- d quotient, only the linearization

$$L(u, v) = \left(\frac{3}{2}u - \frac{1}{2}v, u \right), \quad DL(0) = \begin{pmatrix} 3/2 & -1/2 \\ 1 & 0 \end{pmatrix}$$

survives. Its eigenvalues are 1 and $1/2$, so the degree- d pullback has eigenvalues $1, 1/2, \dots, 2^{-d}$. Table 1 records the five exact blocks without printing an unreadable full matrix.

Table 1: Associated-graded data for the order-four jet.

d	dimension	eigenvalues	trace	determinant
0	1	1	1	1
1	2	1, 1/2	3/2	1/2
2	3	1, 1/2, 1/4	7/4	1/8
3	4	1, 1/2, 1/4, 1/8	15/8	1/64
4	5	1, 1/2, 1/4, 1/8, 1/16	31/16	1/1024

Proposition 1. *On A_4 , eigenvalue 2^{-k} has algebraic multiplicity $5 - k$. Hence*

$$\chi_K(\lambda) = \prod_{k=0}^4 (\lambda - 2^{-k})^{5-k}, \quad \det(I - zK) = \prod_{k=0}^4 (1 - 2^{-k}z)^{5-k}.$$

In particular $\text{tr } K = 129/16$, $\det K = 2^{-20}$, and

$$\text{tr}(K^n) = \sum_{k=0}^4 (5 - k)2^{-kn} \quad (1 \leq n \leq 8).$$

Proof. In the degree-ordered basis, K is block triangular. Its diagonal blocks are the homogeneous pullbacks of L , whose degree- d eigenvalues are the monomials $1^{d-k}(1/2)^k$. Multiplying the five block characteristic polynomials proves both factorizations; summing and multiplying their eigenvalues gives the trace formulas and determinant. $\square$

3 Nonlinear signal and linear control

Let K_{lin} be the pullback obtained by replacing F with L and let $N = K - K_{\text{lin}}$. Independent exact reconstruction finds eleven nonzero entries in N . Every one maps a degree- d column to a row of strictly higher degree, and direct multiplication gives $N^4 = 0$ but $N^3 \neq 0$. Thus $K \neq K_{\text{lin}}$ while their diagonal graded blocks, characteristic polynomials, and finite determinants agree. This comparison prevents two opposite errors: discarding the nonlinear jet, or attributing new finite eigenvalues to a strictly filtration-raising correction.

The canonical ledger also stores the first eight traces of powers. A producer using rational polynomial dictionaries, a checker with a separately written engine, and a fresh SymPy expansion agree on every matrix cell and identity. Canonical replay fixes the evidence bytes; thirteen hostile edits to the model, basis, matrix, invariants, verdict, and nonclaims are all rejected.

4 Route-A boundary

The A1 result is partial-certified only for one fixed point and its order-four local jet. The A2 result is certified-prefix only for the finite fifteen-dimensional quotient A_4 . No global orbit classification, invariant Banach or Hilbert space, global Koopman spectrum, nuclearity, analytic tail theorem, or Fredholm determinant has been established; A3 is not addressed and A4 fails.

In particular, the displayed $\det(I - zK)$ is an ordinary finite matrix determinant. We make no claim about arithmetic/local data, Euler factors, root numbers, automorphy, Riemann zeros, a Hilbert–Pólya operator, or Route B. The release scope is the literal `NO_BAD_EULER_OR_ROOT_NUMBER`.